Investigation into the relationship between the volumetric flow rate of liquid and pipe length/radius

RQ: What is the volumetric flow rate of water with different radii (1mm, 1.5mm, 2mm, 2.5mm, 3mm) and lengths (10cm, 20cm, 30cm, 40cm, 50cm) of an acrylic pipe under constant hydrostatic pressure?

Word count: 2,988

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I. Introduction

1.1 Research question

What is the volumetric flow rate of water with different radii (1mm, 1.5mm, 2mm, 2.5mm, 3mm) and lengths (10cm, 20cm, 30cm, 40cm, 50cm) of an acrylic pipe under constant hydrostatic pressure?

1.2 Introduction

In a local cafe, I have noticed the three types of straws for different drinks. The first one is the default straw used for most liquids, which you can find anywhere. The second had a wider entrance diameter with the same length as that of the first one, used on smoothies. The last one was a straw with a smaller diameter and length used for mixing hot coffee. Upon using all three straws, I have realized that the amount of drink drawn from each straw varied. For instance, when drinking bubble tea, a thick straw pulled a large amount of liquid compared to that of the normal one, while the thin straws exhibited the opposite. Even though this was an obvious phenomenon, it has intrigued me as I was always interested in fluid dynamics, which led me to investigate the volumetric flow rate with different radii and lengths of the pipe under constant water pressure.

This experiment aims to investigate the relationship between the volumetric flow rate of water and the different radii and lengths of the pipe, each under constant water pressure. This will be done by having 5 different pipes with the same radius, but different lengths, and 5 other pipes with the same lengths with varying radii. Therefore, the investigation consists of two parts: part A, investigating the pipe length, and part B, investigating the radius. The volumetric flow rate of water from the pipe will be calculated by measuring the amount of water it lets out in 30 seconds using an electric balance and a stopwatch. Lastly, the constant pressure difference will be controlled by maintaining the same water head by continuously pouring in water to overflow it.

II. Background information

The internal behaviour of a moving liquid is fundamentally explained by its fluid friction, known as dynamic viscosity (μ). Viscosity is a quantitative measure of a fluid's resistance to gradual deformation by shear stress. It represents the internal friction that arises when neighbouring layers of fluid move relative to one another. A higher dynamic viscosity implies greater internal cohesive forces, resulting in a lower flow velocity under a given driving force. Viscosity is highly temperature-dependent. For liquids, an increase in thermal energy lowers the intermolecular cohesive forces, exponentially reducing the fluid's resistance to shear (Britannica Editors, 1998).

Viscosity's effect is quantified by the Reynolds number (Re), a dimensionless ratio of inertial to viscous forces (*What Is Reynolds Number (Re)? (Complete Guide) | SimScale*, 2026), determining whether flow is laminar or turbulent:

$$Re = \frac{2\rho v r}{\mu}$$

Where ρ is fluid density(kg m^{-3}), v is fluid velocity (m s^{-1}), r is the radius of the pipe (m), and μ is dynamic viscosity (Pa s).

Laminar flow occurs at lower velocities, with fluid moving in smooth parallel layers without eddies (*What Is Laminar Flow? Computational Fluid Dynamics | SimScale*, 2026), while turbulent flow occurs at higher velocities with chaotic mixing and swirling velocities that dissipate kinetic energy as heat (*What Is Turbulent Flow? Computational Fluid Dynamics | SimScale*, 2026).

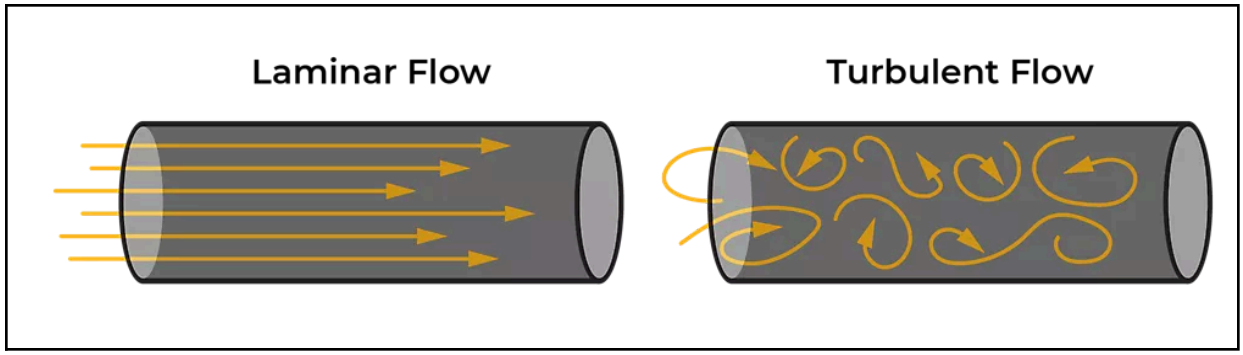


Figure1: The two different regimes of flow (Menter & Menter, 2026)

For a standard cylindrical pipe flowing with water, laminar flow is observed when the Reynolds number is below or equal to 2300 ($Re \leq 2300$), while turbulent flow is observed when the Reynolds number is higher than 4000 ($Re > 4000$). A range between the two is where transitional flow lies($2300 < Re \leq 4000$). In this regime, the flow is inherently unstable and alternates unpredictably between laminar and turbulent flows (Editor Engineeringtoolbox, 2025) . In real life, the turbulent regime is overwhelmingly more common. Therefoe, the research will be focused on the turbulent flow.

2.1 Derivation

The volumetric flow rate (Q) of a fluid passing through a pipe can be expressed as below:

$$Q = v \cdot A = v \cdot (\pi r^2)$$

Where, v is fluid velocity (m s^{-1}), r is the radius of the pipe (m), and A is the cross-sectional area (m^2).

Due to its chaotic behaviour, the turbulent flow exhibits an energy loss due to friction as fluid flows through a pipe. The pressure drop due to friction can be modelled with the Darcy-Weisbach equation (Katemangahas, 2026).

$$\Delta P = f(\frac{L}{2r})(\frac{\rho v^2}{2})$$

Where ΔP is pressure loss (Pa), f is the Darcy friction factor (dimensionless), L is the length of the pipe, r is the radius of the pipe, v is the velocity of the fluid (m s^{-1}), and g is the gravitational acceleration (m s^{-2}).

Rearranging to solve for velocity, it turns to:

$$v = \sqrt{\frac{4gr\Delta P}{fL}}$$

The friction factor can be expressed with the Blasius correlation under the assumptions of using smooth pipes, Newtonian fluids, and turbulent flow. The Blasius correlation shows (Wang et al., 2024):

$$f = \frac{0.316}{Re^{0.25}}$$

Reynolds number can be expressed as:

$$Re = \frac{2\rho v r}{\mu} = \frac{2\rho(\frac{Q}{\pi r^2})r}{\mu} = \frac{2\rho Q}{\pi \mu r}$$

Substituting the Reynolds number leads to:

$$f = \frac{0.316}{(\frac{2\rho Q}{\pi \mu r})^{0.25}}$$

Substituting blasius correlation into the Darcy-Weisbach equation and then the Darcy-Weisbach equation into the volumetric flow rate using velocity, we get:

$$Q = \pi r^2 \sqrt{\frac{4gr\Delta P}{\frac{0.316L}{\left(\frac{2\rho Q}{\pi\mu r}\right)^{0.25}}}}$$

$$Q = \pi r^2 \sqrt{\frac{0.316L\pi^{0.25}\mu^{0.25}r^{0.25}}{2^{0.25}\rho^{0.25}Q^{0.25}}} = \sqrt{\frac{4{\cdot}2^{0.25}gr\Delta P\rho^{0.25}Q^{0.25}}{0.316L\pi^{0.25}\mu^{0.25}r^{0.25}}}$$

$$Q^2 = \pi^2 \frac{4{\cdot}2^{0.25}{\cdot}gr^5\Delta P\rho^{0.25}Q^{0.25}}{0.316L\pi^{0.25}\mu^{0.25}r^{0.25}}$$

$$Q^{1.75} = \pi^2 \frac{4{\cdot}2^{0.25}{\cdot}gr^{4.75}\Delta P\rho^{0.25}}{0.316L\pi^{0.25}\mu^{0.25}}$$

$$Q = \left(\pi^2 \frac{4{\cdot}2^{0.25}{\cdot}gr^{4.75}\Delta P\rho^{0.25}}{0.316L\pi^{0.25}\mu^{0.25}}\right)^{-1.75}$$

$$Q = \left(\pi^2 \frac{4{\cdot}2^{0.25}{\cdot}g\Delta P\rho^{0.25}}{0.316\pi^{0.25}\mu^{0.25}}\right)^{-1.75}\left(\frac{r^{2.71}}{L^{0.571}}\right)$$

$$Q \propto r^{2.71}L^{-0.571}$$

2.2 Hydrostatic pressure

In order to establish a controlled and constant pressure difference (ΔP) across the pipe, the principle of hydrostatic pressure can be utilized through a constant water head. Hydrostatic pressure is the pressure exerted by a fluid at equilibrium at any point in time due to the force of gravity. It is modelled by the equation (*Explaining Hydrostatic and Hydrodynamic Fluid Pressure Components*, n.d.):

$$\Delta P = \rho g h$$

Where g is the acceleration due to gravity (9.81 m s^{-2}) and Δh is the vertical height of the fluid. By implementing a continuous water supply and overflow to maintain the vertical height of fluid, the driving pressure difference across the horizontal pipe is assumed to be constant. It is the primary factor for creating the pressure difference between both ends of the pipe. Thus, by overflowing the tank with water, we can assume the pressure difference becomes constant.

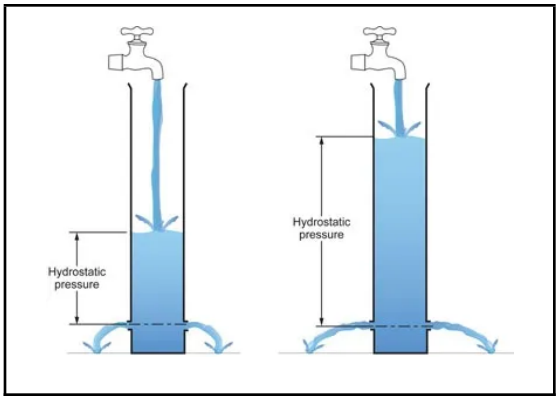


Figure2: Hydrostatic pressure (Mini Physics, 2026)

2.3 Hypothesis

According to the derived equation, increasing the pipe's radius would increase the volumetric flow rate of water by the power of 2.71, and increasing the length of the pipe would decrease the volumetric flow rate of water by the power of -0.571.

III. Methodology

3.1 Justification of method

A specialized custom-built apparatus was required to isolate the effects of pipe length and radius. A custom water tank was built since standard equipment cannot independently vary both pipe length and radius under constant pressure. A continuous-overflow design kept the fluid height (Δh), and thus the driving pressure ($\Delta P = \rho g h$), constant, avoiding the complex time-dependent calculus a falling water would require. Volumetric flow rate ($Q = \frac{V}{t}$) was determined via mass weighing rather than volumetric cylinders, since water's known density allowed instant conversion ($V = \frac{m}{\rho}$), and a graduated cylinder was impractical to wipe dry between every trial.

3.2 Scope of the research

The investigation is divided into two parts:

- Part A: Pipe Length-Volumetric flow rate
- Part B: Pipe Radius-Volumetric flow rate

3.3 Variables

Independent variable (IV): The length/radius of the pipe

- Justification of range: The ranges and intervals for independent variables were chosen for three practical reasons.
    1. The availability of pipes: The minimum intervals of 10cm and 0.5mm for radius were determined by the real-world manufacturing limits of finding small, customizable plastic.
    2. Distinct results: The three specific steps stated above were large enough to produce a distinct, measurable change in the volumetric flow rate (Q) that clearly exceeded the uncertainty.
    3. The upper bound for length: The maximum length of pipe was limited to 50cm. Preliminary testing showed that such a length had internal friction that slowed the water down so much that the hydrostatic pressure could no longer overcome the surface tension, a property of a liquid acting as if it were a stretched elastic sheet, at the exit(Britannica Editors, 2026). Thus, the extension of the water head was imposed.
- Measurement method: The fully intact pipes in the container were measured with a ruler ($\pm$ 0.05 cm)

Dependent variable (DV): The volumetric flow rate of water

- Measurement method: To determine the volumetric flow rate (Q), the mass of the water discharged from the pipe exit was measured over a set timed interval of 30 seconds using a digital stopwatch. Due to the reliance on human reaction, the stopwatch's absolute uncertainty was set to be 0.50s. The recorded mass was then converted to volume by dividing it by the known density. Then, the volume was divided by the time elapsed to calculate it.

$$Q = \frac{m}{\rho \cdot t}$$

Controlled Variables:

Controlled Variable	Explanation	Method of control
Type of fluid used	Different fluids have distinct viscosity and surface tension. Mixing fluids would introduce uncontrolled variables.	Pure tap water from a single source was used for all trials, chosen for its accessibility and consistency given the high volume demands.

Head height of the water	Head height determines hydrostatic driving pressure. Any pressure fluctuation changes flow rate.	Water level was held constant via a continuous-overflow tank. Preliminary tests showed a 5cm head couldn't overcome surface tension in longer pipes, causing dripping instead of flow. The head was therefore fixed at 13cm.
Part A: Radius of the pipe	Part A investigated the length-flow rate relationship, so radius was held constant to isolate the variables.	A fixed radius of 1mm was used for all Part A pipes.
Part B: Length of the pipe	Part B investigated the radius-flow rate relationship, so length was held constant to isolate the variables.	A fixed length of 30cm was used for all Part B pipes.
Material of the pipe	Different materials have varying surface roughness, altering frictional resistance and thus flow rate not being only dependent on the IVs.	Smooth acrylic was used exclusively, minimizing frictional energy loss while remaining easy to customize.
Location of the water inflow source	A source directly above the active pipe would generate localized turbulence, increasing driving pressure.	The water source was placed as far as possible from the pipe under investigation, keeping nearby water calm.

Table1: Controlled variables, explanation, and how it is controlled

3.4 Research Design

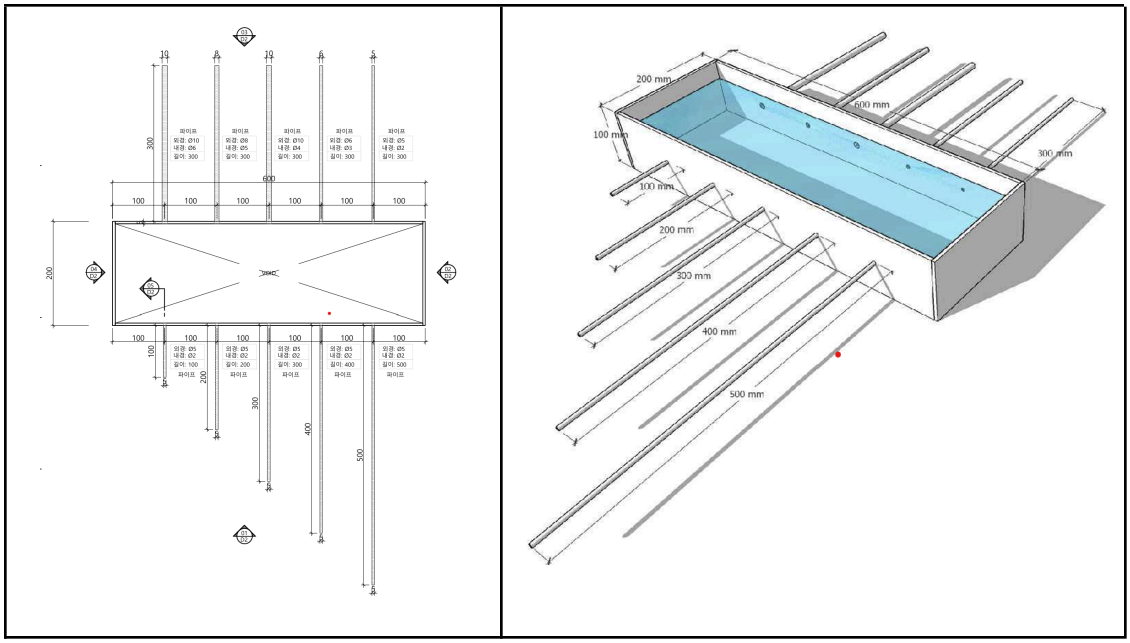


Figure3: Prototype design of the apparatus

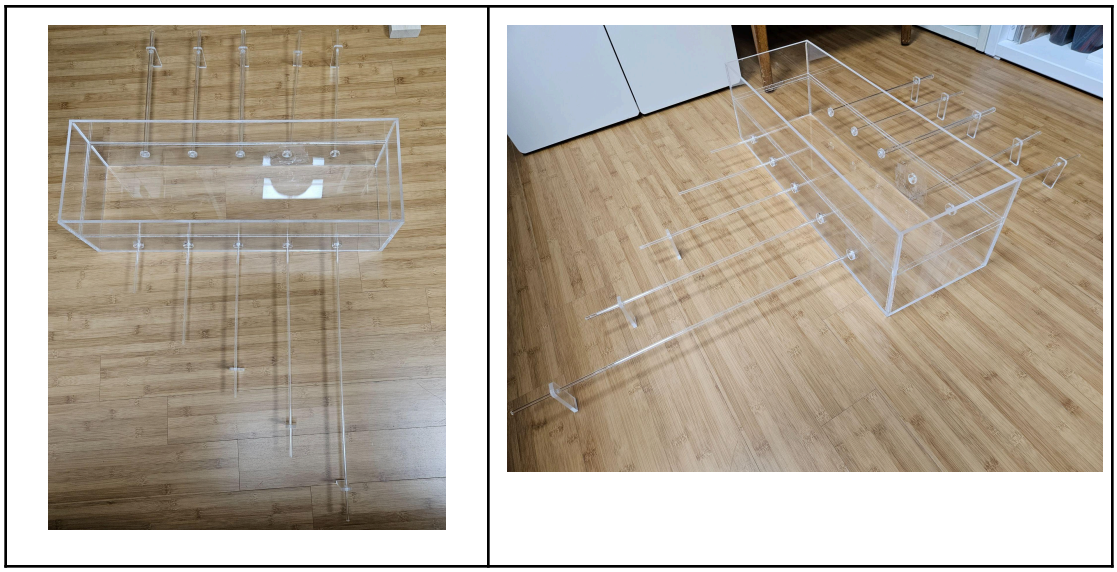


Figure4: The actual apparatus

3.5 Materials and apparatus

- Custom-built water tank
- Tap water
- Digital phone stopwatch ($\pm$0.01s)
- Plate (for collecting discharged water)

- Tape (for blocking non-active pipes)
- Plastic wrap (for blocking non-active pipes)
- Towel (for wiping the used plate after each trial)
- Digital scale (±1g)
- Ruler (±0.05cm)

3.6 Procedure

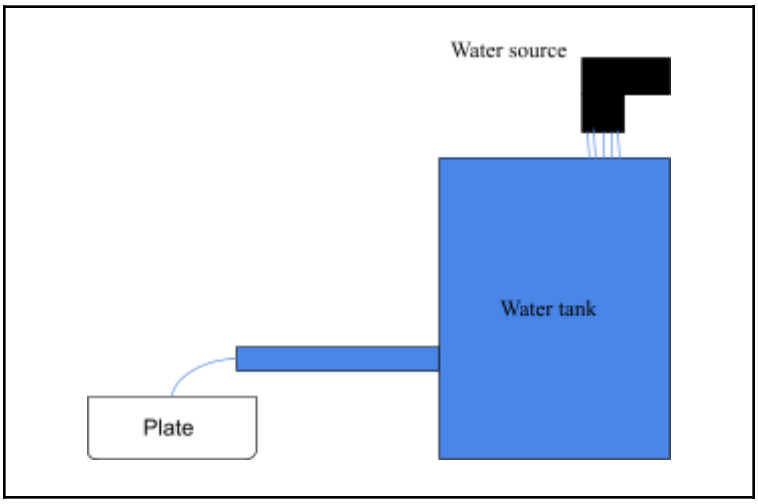


Figure5: Diagram of the experiment setup

1. The source of water was placed to overflow the container
2. The blockage for the desired pipe was removed
3. The source of water was placed as far as possible from the observing pipe
4. The plate's initial weight was measured
5. As the plate was placed under the desired pipe, the stopwatch started
6. As soon as the time hit 30 seconds, the plate was retrieved
7. The plate's final weight was measured
8. The water was disposed of, and the plate was dried with a towel
9. Procedure 2-8 was repeated 5 times
10. The blockage for the pipe was installed
11. The same procedure was repeated for all other pipes

3.7 Risk assessment

Safety concerns: All electronic equipment and digital mass balances were kept isolated from the fluid apparatus to eliminate the risk of short-circuiting or electrical shock.

Ethical concerns: The investigation involved no human or animal subjects, and all data were recorded with scientific integrity.

Environmental concerns: The continuous-overflow design discharged significant water into drainage. To minimize environmental impact, no chemical additives or dyes were used, so only pure water entered the waste stream.

IV. Data

4.1 Part A: Pipe Length-Volumetric flow rate

4.1.1 Raw Data

Length (±0.05cm)	Trials	Initial weight (±1g)	Final weight (±1g)	Time (±0.5s)
10.00	1	56	159	30.2
	2	56	161	30.2
	3	56	161	30.3
	4	56	161	30.2
	5	56	161	30.4
20.00	1	56	138	30.3
	2	56	138	30.3
	3	56	138	30.3
	4	56	138	30.3
	5	56	138	30.4
30.00	1	56	128	30.4
	2	56	129	30.5
	3	56	127	30.3
	4	56	128	30.3

	5	56	127	30.3
40.00	1	56	119	30.4
	2	56	118	30.0
	3	56	117	30.2
	4	56	117	30.3
	5	56	117	30.3
50.00	1	109	169	30.4
	2	109	167	30.2
	3	109	166	30.4
	4	109	165	30.2
	5	109	165	30.2

Table2: Raw data for Part A

Sample calculation for the volumetric flow rate from the 30.00±0.05 cm trial 1, and for the mean of all five trials.

Initial weight: 56±1g (1g-digital scale x1) (2s.f.) Final weight: 128±1g (1g-digital scale x1) (3s.f.) Density (ρ): 1000kg m^{-3}, treated as exact (997kg m^{-3} at 25 °C is a negligible 0.3% difference) Time: 30.4±0.5s (human reaction time, larger than 0.01s-digital stopwatch x1) (3s.f.)	
Calculations	Error Propagation
Change in weight = $Final\ weight\ -\ Initial\ weight$ = 128g-56g = 0.072kg Change in volume = $\frac{Change\ in\ weight}{\rho}$ =	**Absolute uncertainty of change in weight** = $1 + 1 = 2$g **Relative uncertainty of change in weight** = $\frac{2}{72} = 0.0278$

$\frac{0.072}{1000} = 0.000072$ m^3 (2.s.f.) $Volumetric\ flow\ rate = \frac{\Delta Volume}{time} = \frac{0.000072}{30.43}$ =0.00000237 = 2.4×10^{-6} $m^3\ s^{-1}$ (rounded to 2s.f.)	**Relative uncertainty of time** = $\frac{0.50}{30.43} = 0.0164$ **Error propagation** •Relative uncertainty of change in volume = Relative uncertainty (change in weight) = $0.0278 = 0.0278$ •Relative uncertainty of volumetric flow rate = Relative uncertainty (change in volume + time) = 0.0278+0.0164 = 0.0442 **Absolute uncertainty of volumetric flow rate** = $0.0442 \times 2.37 \times 10^{-6} = 1.05 \times 10^{-7} = 0.1 \times 10^{-6}$ (rounded to 1s.f.)
Volumetric flow rate = $(2.4 \pm 0.1) \times 10^{-6}$ $m^3\ s^{-1}$	
Mean of the five trials for L = 30.00cm	
Q of trials 1~5 ($\times 10^{-6} m^3 s^{-1}$): 2.366, 2.393, 2.342, 2.371, 2.345 Mean Q = $\frac{2.366+2.393+2.342+2.371+2.345}{5} \times 10^{-6}$ = 2.363×10^{-6} $m^3 s^{-1}$	ΔQ of trials 1~5 ($\times 10^{-6} m^3 s^{-1}$): 0.1046, 0.1048, 0.1046, 0.1049, 0.1048 Uncertainty of the mean = $\frac{0.1046+0.1048+0.1046+0.1049+0.1048}{5} \times 10^{-6}$ $= 0.1047 \times 10^{-6} = 0.1 \times 10^{-6}$ Half range of Q = $\frac{2.393-2.341}{2} \times 20^{-6}$ =0.026×10^{-6} is smaller than 0.1×10^{-6}, so the averaged uncertainty is used.
Mean volumetric flow rate = $(2.4 \pm 0.1) \times 10^{-6}$ $m^3 s^{-1}$	

Table3: Sample calculation of volumetric flow rate and its error propagation for trial 1 of pipe length 30.00±0.05cm, and of the mean of all five trials.

Therefore, by averaging all trials and their absolute uncertainties, the data for volumetric flow rate and length are as follows.

4.1.2 Processed Data

Length ($\pm$0.05cm)	Average volumetric flow rate ($\times 10^{-6}$ m^3s^{-1})
10.00	3.5$\pm$0.1
20.00	2.7$\pm$0.1
30.00	2.4$\pm$0.1
40.00	2.0$\pm$0.1
50.00	1.9$\pm$0.1

Table4: Processed data for Part A

4.1.3 Graph for Part A

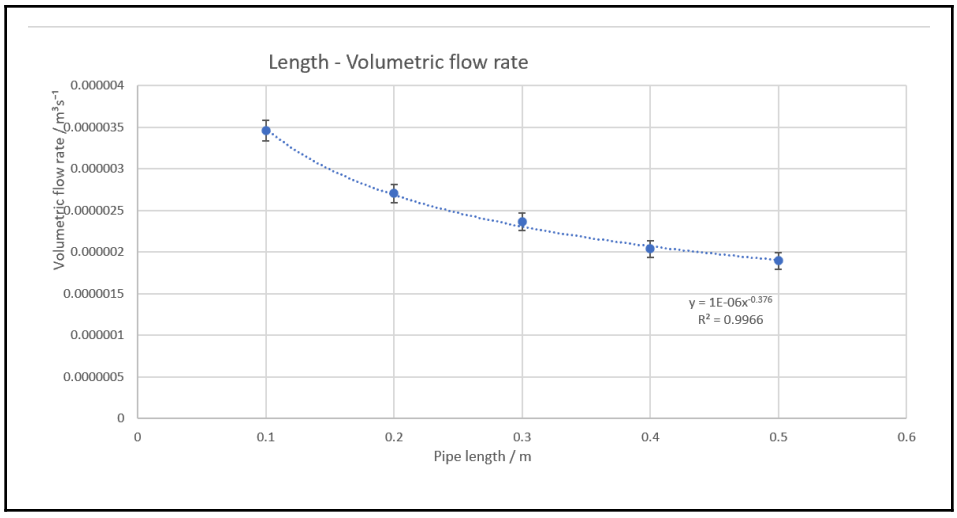


Figure6: Power graph of pipe length-volumetric flow rate with error bars

The horizontal error bars ($\pm\ 0.05cm$) are too small to be visible. The power trendline gives $Q = 1 \times 10^{-6} x^{-0.376}$, so $Q \propto L^{-0.376}$.

4.1.4 Uncertainty of the length exponent

To find the uncertainty of the exponent, $Q = kL^n$ was linearised by taking natural logarithms:

$$ln\, Q = n ln\, L + ln\, k$$

So the gradient of ln Q against ln L equals n. The error bars are

$$\Delta(ln\, A) = \frac{\Delta A}{A}$$

And Δn is half the difference between the steepest and shallowest lines that pass through all error bars (Figure 7).

L / m ($\pm$ 0.0005)	ln (L / m)	$\Delta(ln\, L) = \frac{\Delta L}{L}$	ln(Q / m^3 s^{-1})	$\Delta(ln\, Q) = \frac{\Delta Q}{Q}$
0.1000	-2.303	0.0050	-12.563	0.0286
0.2000	-1.609	0.0025	-12.822	0.0370
0.3000	-1.204	0.0017	-12.940	0.0417
0.4000	-0.916	0.0013	-13.122	0.0500
0.5000	-0.693	0.0010	-13.174	0.0526

Table5: Natural-log data for Part A

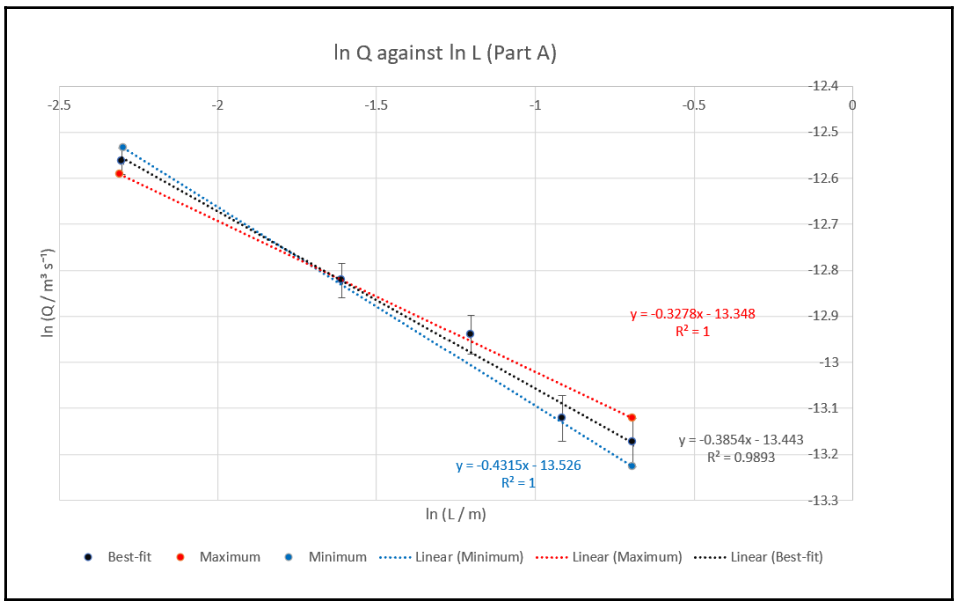


Figure7: ln Q against ln L with best-fit, maximum, and minimum gradient lines

n_{max} = -0.3278, n_{min} = -0.4315

$$\Delta n = \frac{|-0.3278-(-0.4315)|}{2} = 0.0519$$

$$n_L = -0.38 \pm 0.05$$

4.2 Part B: Pipe Radius-Volumetric flow rate

4.2.1 Raw Data

Radius(±0.25mm)	Trials	Initial weight (±1g)	Final weight (±1g)	Time (±0.5s)
1.00	1	109	184	30.3
	2	109	184	30.4
	3	109	183	30.3
	4	109	184	30.3
	5	109	184	30.5
1.50	1	109	323	30.5
	2	109	321	30.4
	3	109	322	30.3
	4	109	325	30.8
	5	109	323	30.3
2.00	1	109	432	30.4
	2	109	434	30.4
	3	109	434	30.5
	4	109	438	30.4
	5	109	435	30.3
2.50	1	109	759	30.5
	2	109	759	30.1

	3	109	759	30.4
	4	109	765	30.5
	5	109	764	30.5
3.00	1	109	1038	30.3
	2	109	1044	30.4
	3	109	1038	30.3
	4	109	1038	30.2
	5	109	1038	30.4

Table6: Raw data for Part B

Sample calculations of volumetric flow rate from 2mm radius trial 1

Initial weight: 109±1g (1g-digital scale x1) (3s.f.)

Final weight: 432±1g (1g-digital scale x1) (3s.f.)

Density (ρ): 1000kg m^{-3}, treated as exact

Time: 30.4±0.5s (human reaction time, larger than 0.01s-stopwatch x1) (3s.f.)

Calculations	Error Propagation
Change in weight = $Final\ weight\ -\ Initial\ weight$ = 432g-109g = 323g = 0.323kg Change in volume = $\frac{Change\ in\ weight}{\rho}$ = $\frac{0.323}{1000} = 0.000323 m^3$ (3s.f.) $Volumetric\ flow\ rate = \frac{\Delta Volume}{time} = \frac{0.000323}{30.37}$ =0.000010635 = $10.6\times 10^{-6} m^3 s^{-1}$(rounded to 3s.f.)	**Absolute uncertainty of change in weight** = $1 + 1 = 2$g **Relative uncertainty of change in weight** = $\frac{2}{323} = 0.00619$ **Relative uncertainty of time** = $\frac{0.50}{30.37} = 0.0165$ **Error propagation** •Relative uncertainty of change in volume = Relative uncertainty (change in weight) =

	$0.00619 = 0.00619$ •Relative uncertainty of volumetric flow rate = Relative uncertainty (change in volume + time) = 0.00619+0.0165 = 0.02269 **Absolute uncertainty of volumetric flow rate** = 0.02269×0.000010635 = 0.000000241 = 0.2×10^{-6}(rounded to 1s.f.)
Volumetric flow rate = $(10.6 \pm 0.2)\times 10^{-6}$ m^3s^{-1}	

Table7: Sample calculations for volumetric flow rate and its error propagation for trial 1 of pipe radius 2.0 ±0.25mm

As in Part A, each mean and its uncertainty were obtained by averaging the five trials.

4.2.2 Processed Data

Radius (±0.25mm)	Average volumetric flow rate ($\times 10^{-6}$ m^3s^{-1})
1.00	2.5±0.1
1.50	7.0±0.2
2.00	10.6±0.2
2.50	21.5±0.4
3.00	30.7±0.6

Table8: Processed data for Part B

4.2.3 Graph for Part B

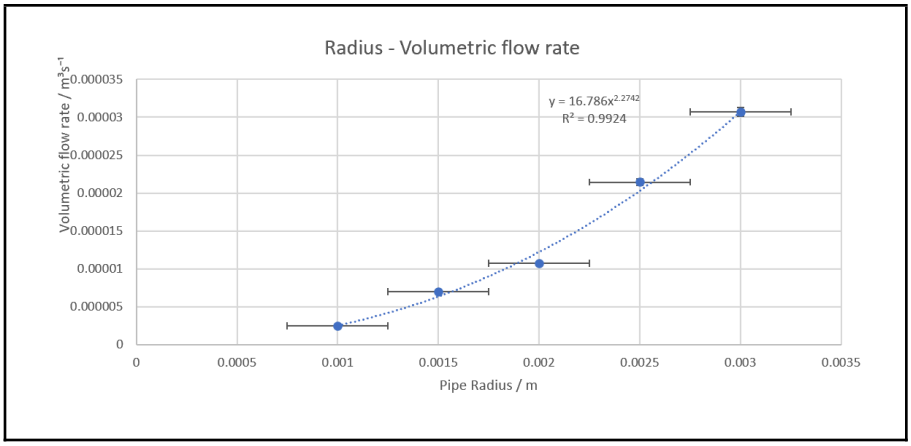


Figure8: Power graph of pipe radius-volumetric flow rate with error bars

The horizontal bars ($\pm\ 0.25mm$) are much larger than the vertical ones. The power trendline gives $Q = 16.786r^{2.2742}$, so $Q \propto r^{2.2742}$.

4.2.4 Uncertainty of the radius exponent

The same method, applied to ln Q against ln r, gives Table 9 and Figure 9.

r / m($\pm\ 0.00025$)	ln (r / m)	$\Delta(\ln r) = \frac{\Delta r}{r}$	ln (Q / m^3s^{-1})	$\Delta(\ln Q) = \frac{\Delta Q}{Q}$
0.00100	-6.908	0.250	-12.899	0.0400
0.00150	-6.502	0.167	-11.870	0.0286
0.00200	-6.215	0.125	-11.455	0.0189
0.00250	-5.991	0.100	-10.747	0.0186
0.00300	-5.809	0.083	-10.391	0.0195

Table9: Natural-log data for Part B

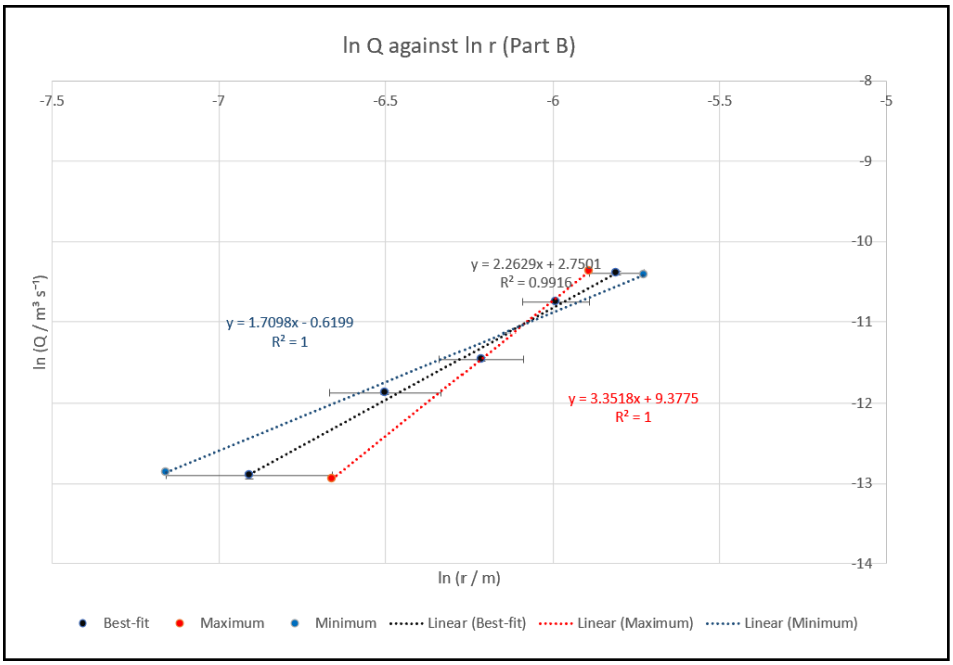


Figure9: ln Q against ln r with the best-fit, maximum, and minimum-gradient lines

$n_{max} = 3.3518, \qquad n_{min} = 1.7098$

$$\Delta n = \frac{|3.3518 - 1.7098|}{2} = 0.821$$

$$n_r = 2.3 \pm 0.8$$

4.3 Analysis

In order to determine the joint impact of pipe length and radius on volumetric flow rate, the individual power-law relationships were combined. From Part A and Part B, we know that:

$$Q \propto L^{-0.376}$$

$$Q \propto r^{2.2742}$$

$$\rightarrow Q \propto \frac{r^{2.2742}}{L^{0.376}}$$

$$Q = k \cdot \frac{r^{2.2742}}{L^{0.376}}$$

Constant k was determined by plotting the volumetric flow rate against the scaling factor of $\frac{r^{2.2742}}{L^{0.376}}$ for both Part A and Part B. In Part A, r would be a constant of 0.001, and in Part B, L would be a constant of 0.30m. By plotting its value against volumetric flow rate, we get the following graph with its best-fit line.

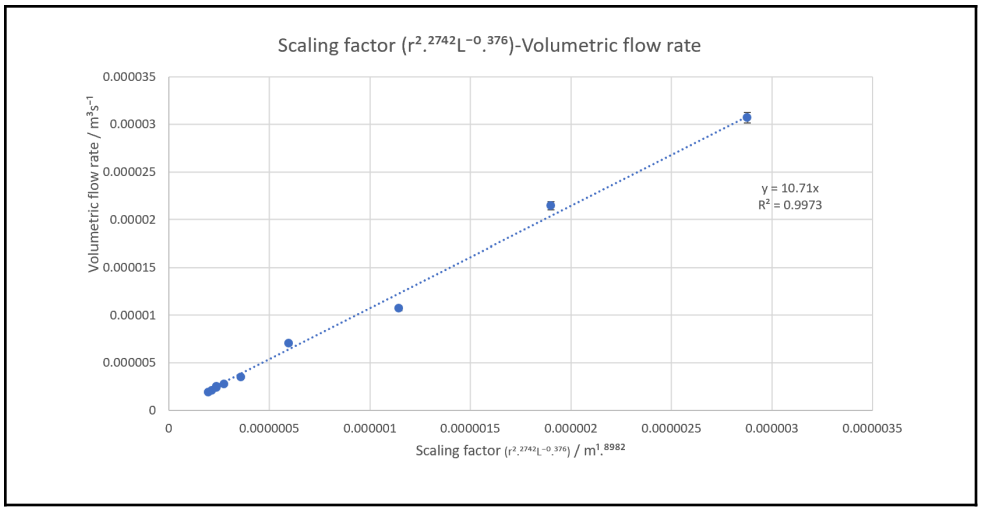


Figure10: linear graph of scaling factor-volumetric flow rate

The final gradient is 10.71 with a high coefficient of determination of 0.9973. Therefore, the final relationship of volumetric flow rate and pipe length/radius with the obtained k and the exponent uncertainties would be:

$$Q = 10.71 \frac{r^{2.3\pm0.8}}{L^{0.38\pm0.05}}$$

V. Conclusion

This investigation aimed to determine the joint empirical relationship between the volumetric flow rate (Q) of water through a pipe and its geometric dimensions, radius (r) and length (L). As evidenced by Figure6 and Figure8, the results partially align with the initial hypothesis: increasing the radius increased the volumetric flow rate in a power relationship, while increasing the length decreased the volumetric flow rate as a negative power. By synthesizing the power-law relationships extracted from experimental data, the final model was determined to be:

$$Q = 10.71 \frac{r^{2.3\pm0.8}}{L^{0.38\pm0.05}}$$

Physically, a longer pipe lowers Q because viscous friction along the walls dissipates energy over a longer path, while a wider pipe raises Q because the cross-section grows and a smaller fraction of the fluid lies in the slow layer near the walls.

Table 10 compares the experimental exponents with theoretical values. A difference is significant if the theoretical value lies outside the experimental uncertainty range.

Exponent	Experimental value (range)	Theoretical value	Is it inside the range?	Significant?
Length, n_L	-0.38 ± 0.05 (-0.43~-0.33)	-0.571	No	Yes
Radius, n_r	2.3 ± 0.8 (1.5~3.1)	2.71	Yes	No

Table10: Comparing experimental value and theoretical value

The experimental value (-0.38 ± 0.05) for length exponent lies outside the theoretical prediction (-0.571), so the length deviation is significant and cannot be attributed to random uncertainty. For radius, the theoretical value (2.71) lies within the experimental value (2.3 ± 0.8), so the difference is not significant. However, the Reynolds numbers calculated later on show that most flows were laminar or transitional, so the turbulent model is not appropriate for comparison. Comparing this with the laminar Hagen-Poiseuille ($Q \propto r^4 L^{-1}$), both exponents differ significantly (*Poiseuille's Law | Harvard Natural Sciences Lecture Demonstrations*, n.d.).

VI. Evaluation

Even though the model explicitly demonstrates a non-linear relationship where fluid flow remains positively correlated with pipe radius and inversely correlated with pipe length, the specific values of exponents heavily deviate from the derived equation under the assumptions of a smooth pipe, Newtonian fluid, and turbulent flow:

$$\text{Theoretical: } Q = \left(\pi^2 \frac{4 \cdot 2^{0.25} \cdot g \Delta P \rho^{0.25}}{0.316 \pi^{0.25} \mu^{0.25}}\right)^{-1.75} \left(\frac{r^{2.71}}{L^{0.571}}\right)$$

$$\text{Experimental: } Q = 10.71 \frac{r^{2.3 \pm 0.8}}{L^{0.38 \pm 0.05}}$$

In order to understand the cause of the significant length deviation, the Reynolds number (Re) was calculated for all flows through the pipe.

Part A (radius = 1.00±0.25mm)

Pipe length (±0.0005m)	Reynolds number (Re)

Part B (length = 30.00±0.05cm)

Pipe radius (±0.0005m)	Reynolds number

0.1000	2197.385		0.0010	1565.981
0.2000	1719.091		0.0015	2974.81
0.3000	1501.554		0.0020	3403.131
0.4000	1294.919		0.0025	5451.299
0.5000	1204.909		0.0030	6501.679

Table11: Calculated Reynolds number for all trials(dynamic viscosity = 0.001 Pa s used)

The calculation of Reynolds numbers reveals a regime inconsistency between the two parts of the experiment. In Part A, where the pipe radius was held constant at 1mm, the fluid remained entirely bounded within the laminar flow (Re<2300). Conversely, Part B captured a clear, inconsistent flow regime. The smallest radius exhibited laminar flow, the intermediate radii (1.5mm and 2mm) fell within the transitional flow, and the largest radii (2.5mm and 3mm) achieved fully developed turbulent flow (Re>4000). This multi-regime distribution directly challenges the underlying assumption for the derived Darcy-Weisbach and Blasius equations, which assume uniform, fully developed turbulence.

6.1 Major Limitations from the apparatus

The uniform laminar behaviour observed across all trials in Part A, contrasted with the multi-regime in Part B, reveals a geometric constraint in the apparatus design. The Reynolds number equation, $Re = \frac{2\rho Q}{\pi \mu r}$, dictates that the flow regime is heavily dependent on the ratio of the volumetric flow rate to the pipe radius, as all other variables are practically constant.

In part A, the radius was locked at a small boundary (r = 1mm). Because the internal cross-sectional area was so small ($A = 3.14 \times 10^{-6}$), the narrow boundary walls exerted a dominating viscous drag across the entire fluid flow. This severe spatial restriction limited the volumetric flow rate. Consequently, even at the shortest pipe where friction was minimized, the fluid momentum was suppressed by the narrow pipe boundaries, limiting the Reynolds number below the critical threshold (Re = 2300) and preserving consistent laminar flow. Thus, the data should theoretically be evaluated against the Hagen-Poiseuille law, which expresses an inverse proportionality between pipe length and volumetric flow rate ($Q = r^4 L^{-1}$) (*Poiseuille's Law | Harvard Natural Sciences Lecture Demonstrations*, n.d.). But the experiment yielded an exponent of -0.38 ± 0.05. This significant deviation suggests that it may not have been fully developed laminar flow. Instead, most of the pipe flow was likely dominated by an

entrance length effect, where the velocity profile was still forming, altering the volumetric flow rate (*University of Sydney, 2005*).

In Part B, however, the length was held constant at 30cm while the radius expanded up to 3mm. Expanding the radius increases the cross-sectional area by a power of two ($A \propto r^2$), which exponentially reduced the proportion of fluid directly in contact with the friction-inducing internal pipe walls. This allowed the volumetric flow rate to surge up. Because Q scales far more aggressively than r, the $\frac{Q}{r}$ ratio grew rapidly. As a result, the fluid's inertial forces completely overtook the internal viscous friction, forcing the system out of its laminar state, through transitional flow, and directly to turbulent flow.

Therefore, to guarantee a fully developed turbulent regime across all ten pipes, future design must utilize a larger minimum pipe radius. Mathematically expanding the radius increases the $\frac{Q}{r}$ ratio in $Re = \frac{2\rho Q}{\pi \mu r}$, driving the minimum Reynolds number to be past the Re = 4000 turbulent threshold for every trial.

6.2 Other Limitations

Apart from the major limitations from the research design, there were some other limitations found during the experiment.

Limitations	Explanation	Improvements
Minor pressure losses from pipe inlet	As water enters the tiny pipe inlet, the sudden contraction at the sharp corner creates localized turbulence and energy loss, decreasing flow velocity. This could lower the radius exponent, although its difference from theory was not significant.	Chamfer the inlet into a funnel shape rather than a sharp 90-degree cut to minimize contraction losses.
The temperature of the water	Despite constant room temperature, incoming tap water temperature fluctuates unpredictably, altering viscosity and thus fluid velocity, which the model assumes fixed.	Monitor the tank with a real-time thermometer probe, and pre-draw water into a storage container to match ambient temperature before trials.

Surface tension	Water's high surface tension caused it to cling to the pipe's outer rim rather than flow smoothly, acting as an unintended resistive force against the hydrostatic pressure, worsening with narrower pipes.	Apply a hydrophobic coating to the pipe exits' exterior rim to lower surface tension and prevent clinging.
High relative uncertainty in radius	The $\pm$0.25mm absolute uncertainty introduces up to 25% relative uncertainty given the smaller radii (1.0- 3.0 mm), so the regression line may differ significantly from the true relationship.	Measure the internal radius using digital calipers ($\pm\ 0.01mm$)

Table12: Limitations, explanation, and improvements for the investigation

6.3 Extension

The project could be improved by taking everything mentioned in the evaluation section into account and achieving a consistent turbulent flow for all trials. Moreover, since the development of flow regimes was understood with $\frac{Q}{r}$ the ratio, the research could be extended to analyzing the other flow regime, which is laminar. The laminar investigation could be designed by carefully calculating the entrance length for pipes to develop a full laminar flow, so the huge deviation observed from Part A doesn't occur again.

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